

ANABAR, Russian Federation

WMO: 216080

Lat: 73.217N

Lon: 113.500E

Elev: 44

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-51.2	-48.5	-53.0	0.2	-50.7	-50.4	0.3	-48.2	34.3	-5.2	31.8	-12.2	8.3	120	1.103

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	66.3	60.3	62.5	56.9	58.5	54.4	60.8	65.6	57.6	61.9	54.9	58.2	12.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.1	72.3	64.6	55.2	65.0	60.2	52.5	58.7	57.6	27.0	65.4	24.9	61.9	23.2	58.1	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.0	27.8	24.4	DB	-53.3	72.4	3.5	5.4	-55.8	76.3	-57.9	79.5	-59.8	82.6	-62.4	86.5	(4)
(5)			WB	-53.3	64.2	3.5	4.8	-55.8	67.6	-57.9	70.4	-59.8	73.1	-62.4	76.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-27.0	-28.5	-15.3	1.0	18.1	40.2	47.6	46.8	36.0	13.8	-11.6	-21.8		
	DBStd	30.12	13.98	13.02	15.19	11.08	9.41	8.38	7.32	6.55	6.16	11.16	13.28	12.24		
	HDD50	15279	2386	2197	2023	1471	989	316	134	142	423	1123	1848	2227		
	HDD65	20627	2851	2617	2488	1921	1454	743	541	563	871	1588	2298	2692		
	CDD50	129	0	0	0	0	0	23	61	43	2	0	0	0		
	CDD65	4	0	0	0	0	0	1	3	0	0	0	0	0		
	CDH74	26	0	0	0	0	0	2	24	0	0	0	0	0		
	CDH80	2	0	0	0	0	0	0	2	0	0	0	0	0		
	WSAvg	11.9	10.9	10.5	11.2	12.5	12.7	12.7	12.6	12.6	12.2	11.0	11.8	11.6		
	PrecAvg	11.1	0.7	0.5	0.6	0.6	0.9	0.8	1.2	1.6	1.4	1.3	0.9	0.7		
	PrecMax	15.6	2.1	1.1	1.0	1.6	1.9	1.7	2.5	3.1	3.4	2.3	1.7	1.5		
(15) Precipitation	PrecMin	6.4	0.1	0.0	0.1	0.0	0.2	0.2	0.4	0.6	0.4	0.4	0.3	0.3		
	PrecStd	2.0	0.4	0.2	0.2	0.4	0.4	0.4	0.5	0.7	0.7	0.4	0.4	0.3		
	DB	23.8	6.9	26.1	32.7	39.5	70.7	78.4	68.3	56.5	34.2	26.8	16.2			
	MCWB	23.7	6.2	25.7	30.6	36.9	62.4	68.5	61.2	53.6	33.8	26.3	15.9			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	4.4	2.8	21.7	27.5	35.7	64.2	68.5	64.4	50.2	32.3	20.8	5.9			
	2% MCWB	4.2	2.6	20.9	26.3	33.8	59.0	61.8	59.3	48.5	31.8	20.6	5.6			
	5% DB	-0.4	-4.0	17.1	23.2	34.3	58.0	63.0	61.1	46.8	29.7	12.9	1.0			
	5% MCWB	-0.5	-4.3	16.7	22.4	32.7	53.3	57.3	55.9	45.3	29.2	12.5	0.8			
	10% DB	-6.6	-8.7	9.8	18.9	32.2	52.6	59.1	57.2	43.7	27.4	7.3	-3.6			
	10% MCWB	-6.9	-8.9	9.2	18.2	31.1	49.3	54.8	54.0	42.5	26.9	6.8	-3.9			
	0.4% WB	23.5	6.3	25.7	30.5	37.7	63.3	67.9	62.4	55.5	33.9	26.7	16.0			
	0.4% MCDB	23.7	6.7	26.1	32.4	39.0	68.1	77.5	67.3	56.8	34.0	26.7	16.3			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	4.3	2.5	21.0	26.3	34.2	58.3	61.9	59.4	48.7	31.8	20.5	5.7			
	2% MCDB	4.4	2.8	21.7	27.2	35.4	63.8	67.1	64.1	49.5	32.2	20.8	5.9			
	5% WB	-0.5	-4.2	16.6	22.5	32.7	53.6	58.1	56.7	45.6	29.3	12.7	0.8			
	5% MCDB	-0.4	-4.0	17.1	23.2	34.0	57.7	62.6	60.2	47.2	29.6	13.1	1.0			
	10% WB	-6.8	-8.9	9.4	18.3	31.1	49.3	55.2	54.2	42.5	26.9	6.9	-3.8			
	10% MCDB	-6.6	-8.8	9.8	18.9	32.3	52.5	58.9	57.3	43.4	27.4	7.3	-3.4			
	MDBR	11.1	11.1	13.3	15.5	10.2	10.2	11.8	10.9	6.7	9.7	11.0	10.7			
	5% DB	16.0	18.1	18.3	19.0	11.1	20.2	17.5	17.2	12.9	9.4	15.6	15.7			
(36) Mean Daily Temperature Range	MCWBR	15.9	17.9	17.9	18.8	10.1	15.5	12.3	12.5	10.9	9.2	15.5	15.2			
	MCDBR	16.0	17.4	18.3	19.7	11.0	20.4	17.5	16.5	13.1	9.2	15.9	15.1			
	MCWBR	15.8	17.1	17.9	19.4	10.2	15.9	12.8	12.9	11.2	9.2	15.8	15.0			
	MCDBR	15.8	17.1	17.9	19.4	10.2	15.9	12.8	12.9	11.2	9.2	15.8	15.0			
(40) Clear-Sky Solar Irradiance	taub	N/A	0.184	0.278	0.309	0.335	0.341	0.372	0.369	0.297	0.193	N/A	N/A			
	taud	N/A	1.901	2.020	2.063	2.105	2.359	2.341	2.310	2.394	1.946	N/A	N/A			
	Ebn at Noon	0	119	203	247	258	264	248	228	211	104	0	0			
	Edh at Noon	0	9	23	34	38	31	30	27	17	7	0	0			
(44) All-Sky Solar Radiation	RadAvg	0	42	473	1275	1858	1827	1546	928	409	92	0	0			
	RadStd	0	4	22	55	138	154	111	127	47	13	0	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page